



Active Transportation & the Built Environment

Accessibility, Design, and Planning

Fall 2026

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City & Regional Planning

A researcher who likes to code

- Background: urban systems, computer science, & economics
- MIT DUSP , NYU CS (VIDA lab)
- Open-source developer
- Research: Accessibility, computer vision, urban mobility, pedestrian safety, and heat exposure
- Designing interactive visual analytics tools



Berkeley
Boston
New York
Warsaw
Tehran

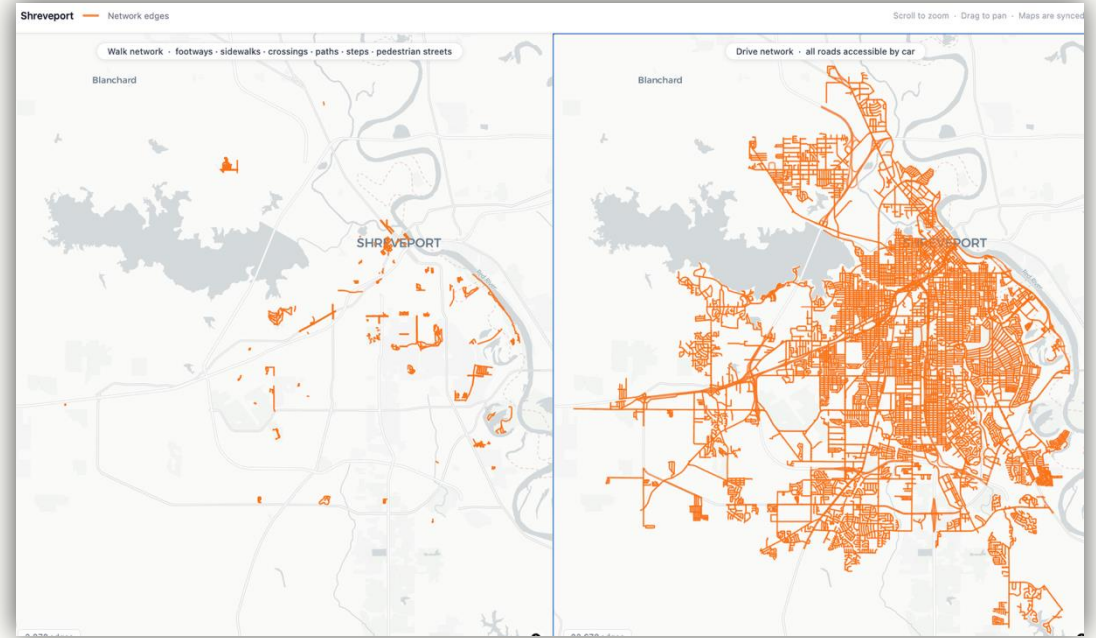
SHREVEPORT, LOUISIANA — 2014

Ticketed for rolling on the road; the sidewalk was impassable!



Cedric Murphy has to balance himself on the sidewalk at the corner of Olive and Dillingham where he received a ticket for riding his wheelchair in the street.

Henrietta Wildsmith/The Times



12 years later, left: pedestrian network, right road network

**SIDEWALK CLOSED
PEDESTRIANS
CROSS OVER**



COLORADO
BARRECADE CO.

COLORADO BARRECADE 303-822-785



120%

measured difference in pedestrian connectivity
when using road centerlines instead of
sidewalk networks, in the same neighborhood.

(Chin et al., 2008)

PUBLIC HEALTH

Physical activity, obesity, cardiovascular disease modeled on walkability indices that assume sidewalks exist where they do not.

TRANSPORTATION PLANNING

Transit access, first/last mile connectivity, routing for people with disabilities, all distorted by absent pedestrian network data.

URBAN ECONOMICS

Property values, job access, investment prioritization, built on walkability scores with no sidewalk ground truth.

65%

of all pedestrian deaths in 2023
occurred at locations **without a sidewalk**

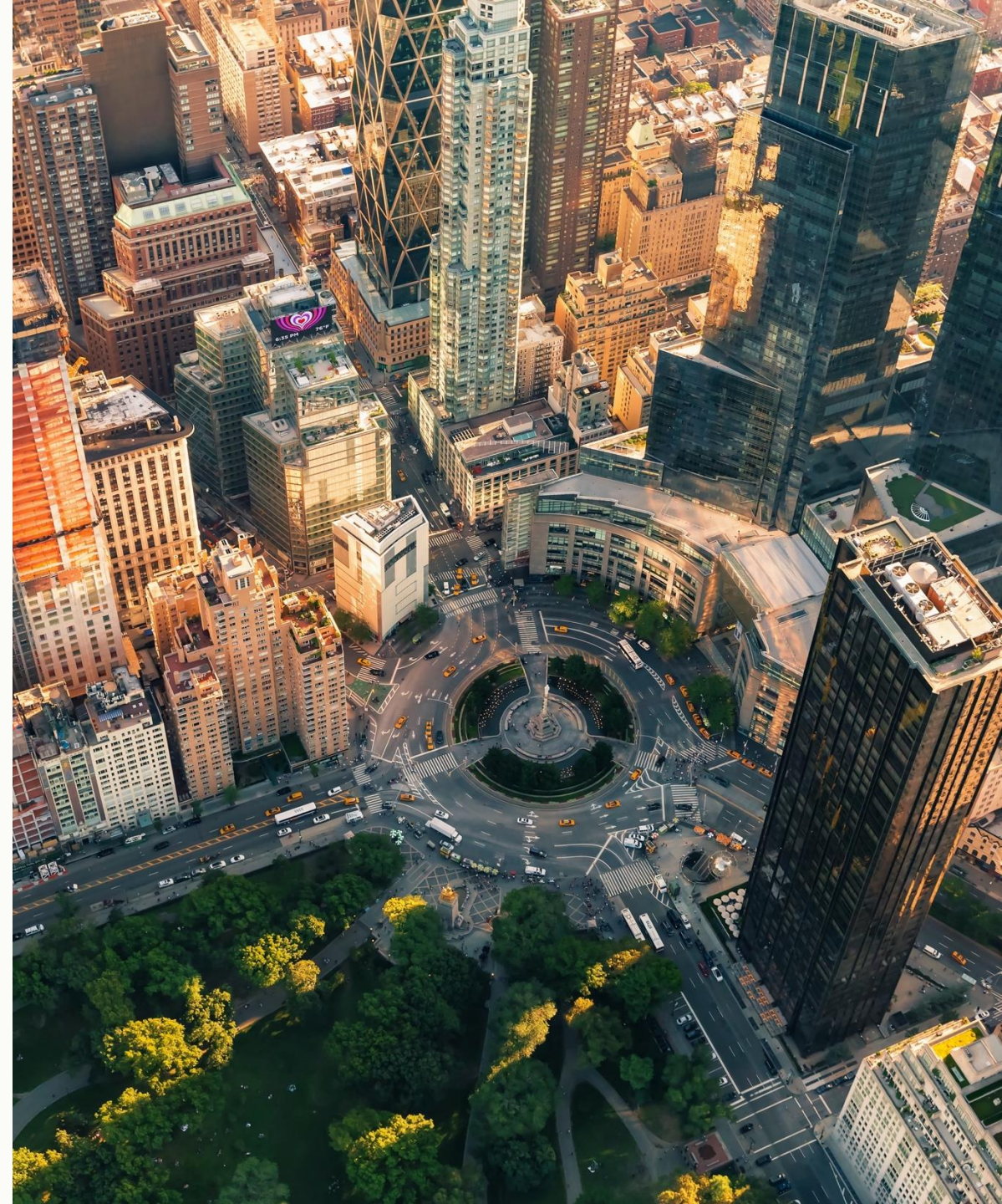
+1,164

more pedestrian deaths **in no-sidewalk
locations** since 2017, compared to only
+167 in locations that have sidewalks.



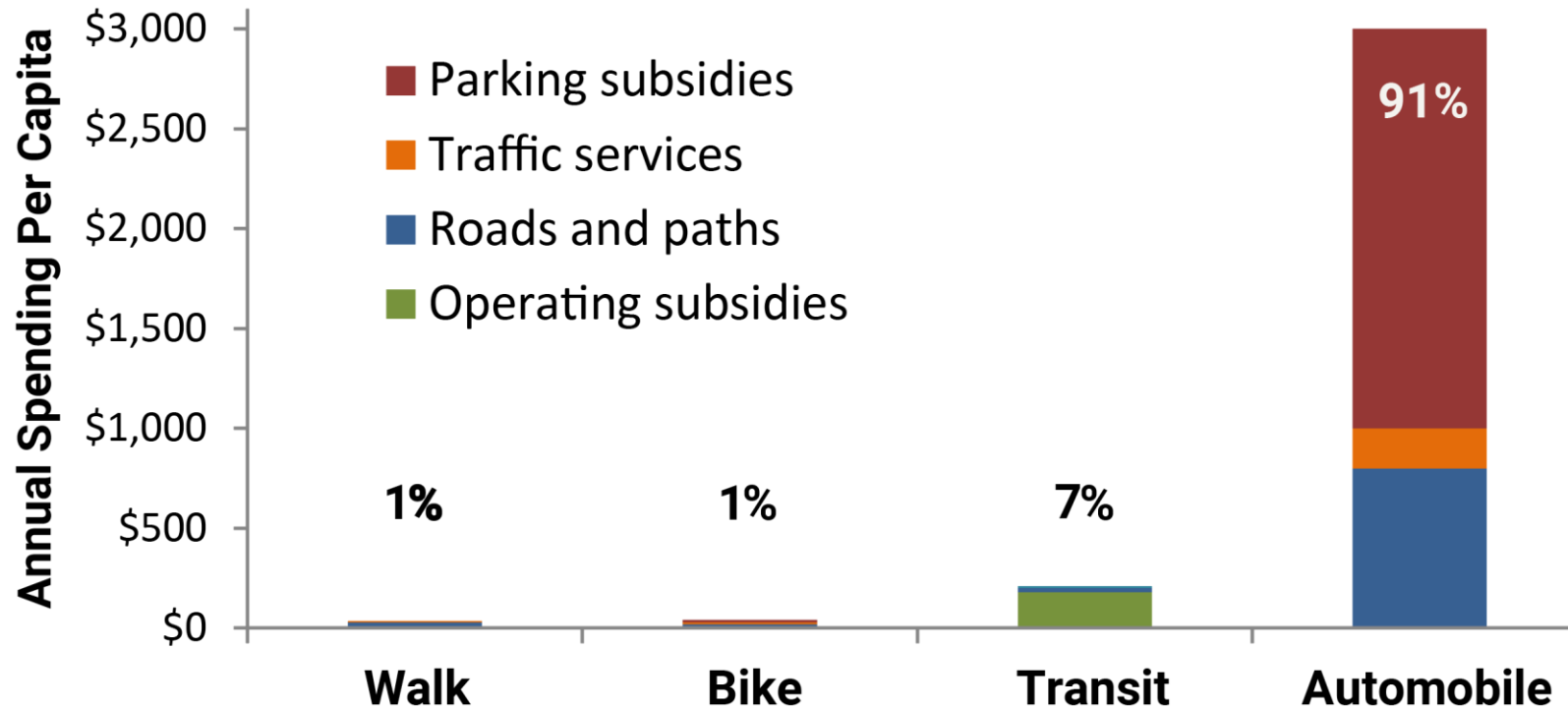
Data Availability

- Clear inequities in what data we have and whose interest it represents
- Data availability drives where decisions are focused



A third of US population are non-drivers

Infrastructure Spending (APTA 2020; FHWA 2018; LAB 2018; Litman 2020)





Cynde Soto @cynde_soto · May 11

It's time for Los Angeles to Stop Trippin' and fix its sidewalks, crosswalks and access ramps. As a wheelchair user my independence depends on it!

@BobBlumenfield @CD12LA @kdeleon @MitchOFarrell @HerbJWesson

#LASidewalks

44
million
individuals
with
Disability
in the US

7.6
million
adults
with
Disability
in CA

3.3
million
adults
with
Mobility
Disability
In CA

disabilitystatistics.org

cdc.gov



**Any comprehensive analysis
of pedestrian infrastructure
must include a threefold understanding of**



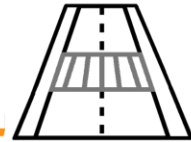
Where

sidewalks are



How

they are connected



What

their condition is

Lack of Data

Among a sample of 178 municipalities across the U.S. :

Only 20% had sidewalk data

Only 11% had crosswalk data

Only 8% had sidewalk material data

(Dietz, 2021)



U.S. Department of Transportation

DOT Issues Final Rule Establishing Accessibility Standards for Pedestrian Facilities in the Public Right-of-Way

Wednesday, December 18, 2024

PROWAG 2024

Federal accessibility standards for pedestrian infrastructure are now binding and enforceable.

DOT's December 2024 final rule adopts Public Rights-of-Way Accessibility Guidelines as federal standards. To comply, cities must first know what exists. Most have never assessed their own sidewalks.

IIJA 2021

\$1.2 trillion in federal infrastructure funding, with ADA compliance as a condition of eligibility.

The Bipartisan Infrastructure Law ties active transportation grants to accessibility compliance. Cities without sidewalk inventories cannot compete for available funds. First violation: \$75,000 fine. Subsequent: \$150,000.

**The mandate has arrived.
The data hasn't.**





Davis

Walk network
Footways, paths, pedestrian streets

Road network walkable on foot
Residential + primary

Berkeley

Walk network
Footways, paths, pedestrian streets

Road network walkable on foot
Residential + primary

Merced

Walk network
Footways, paths, pedestrian streets

Road network walkable on foot
Residential + primary

Santa Barbara

Walk network
Footways, paths, pedestrian streets

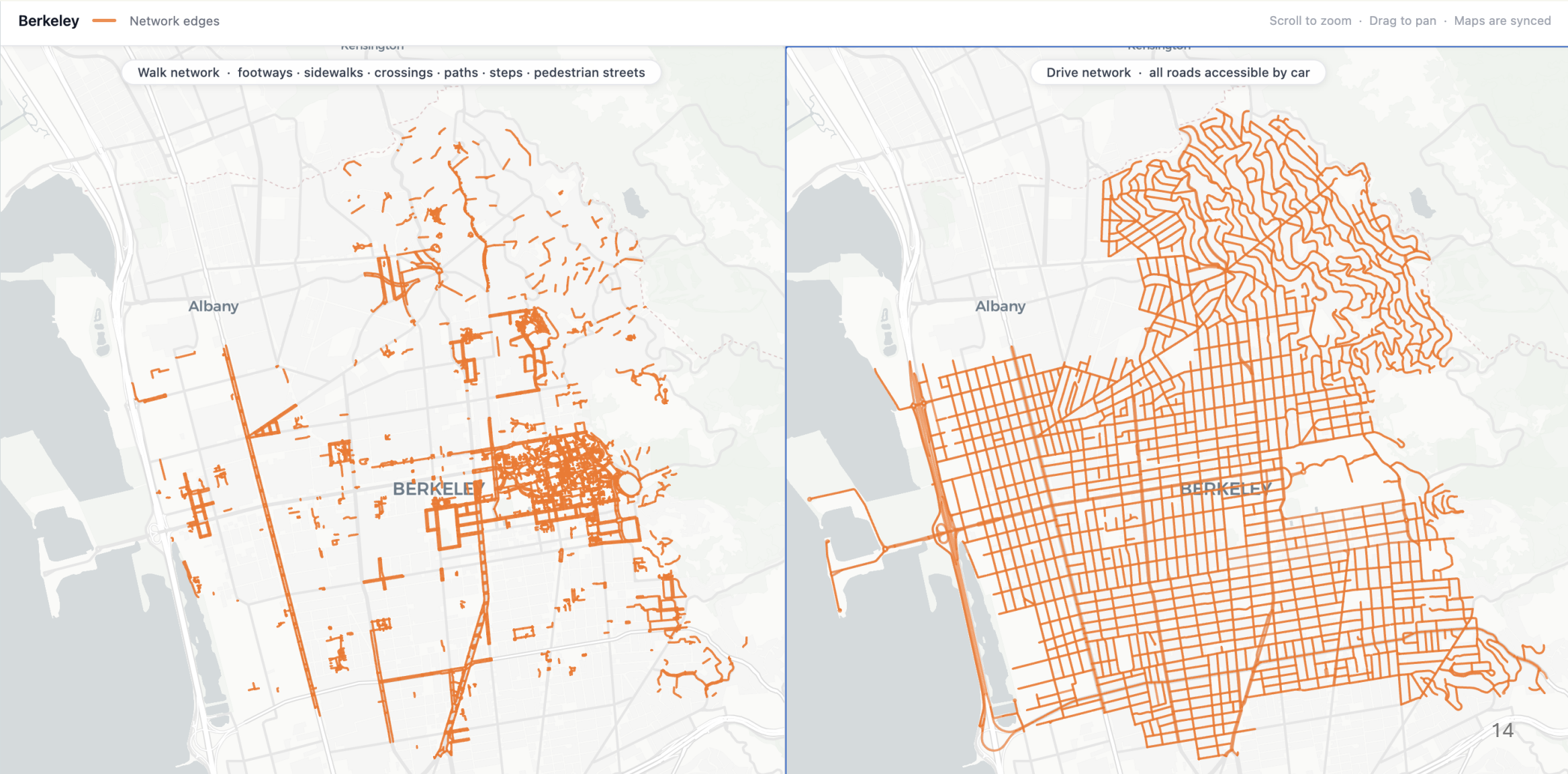
Road network walkable on foot
Residential + primary

What about OSM?

Berkeley

Walk Network

Road Network



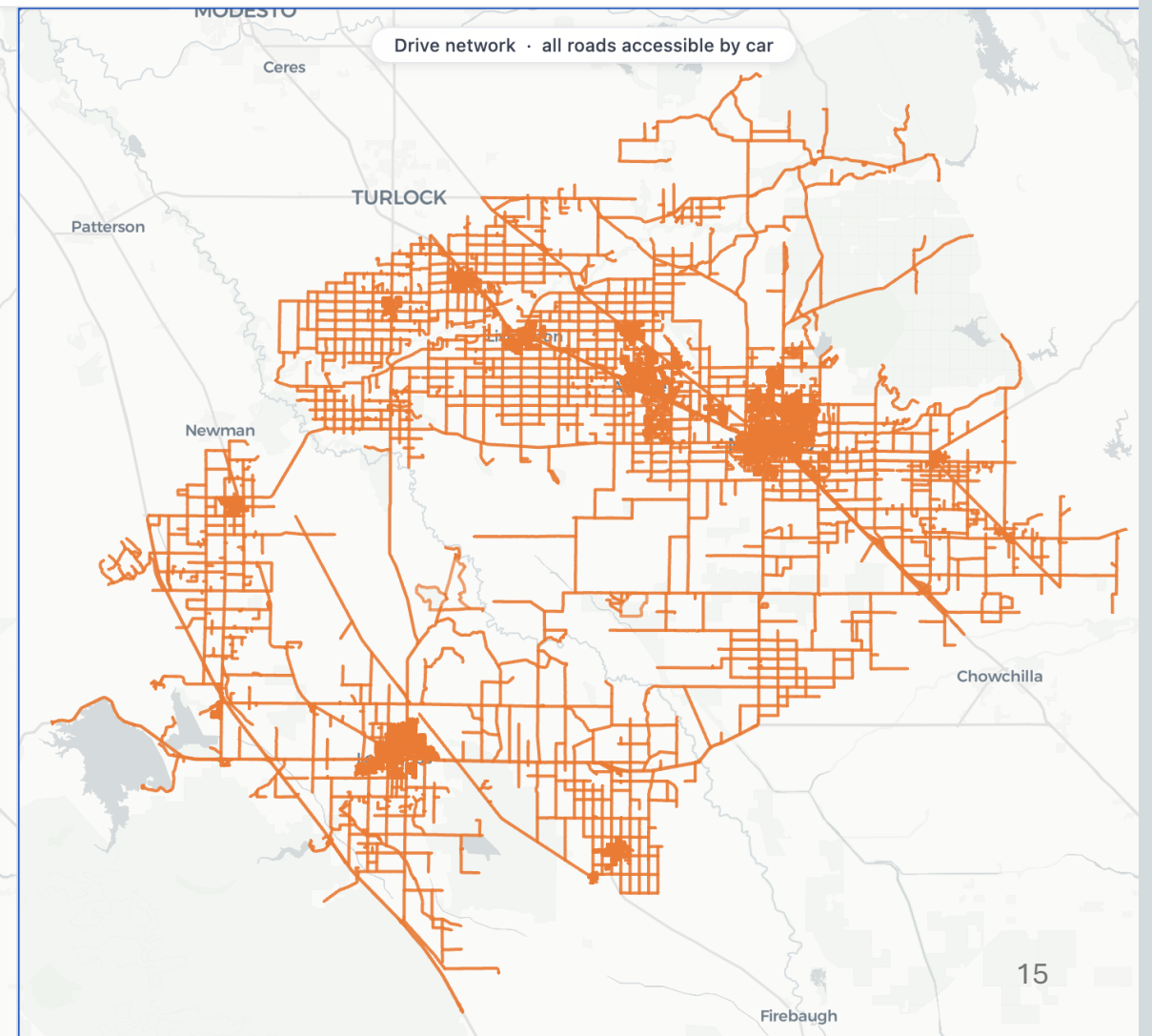
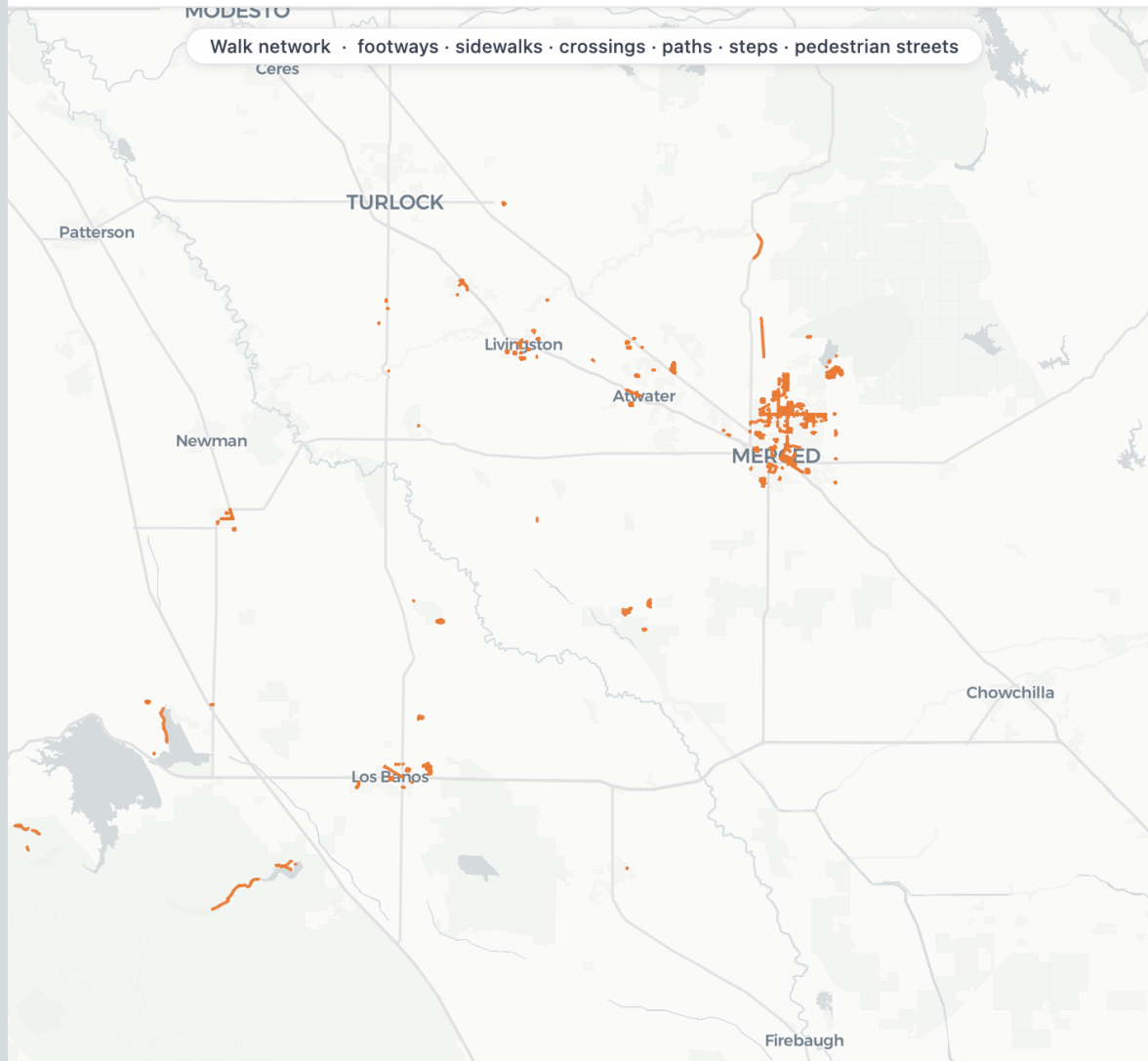
Mercede

Walk Network

Road Network

Merced — Network edges

Scroll to zoom · Drag to pan · Maps are synced



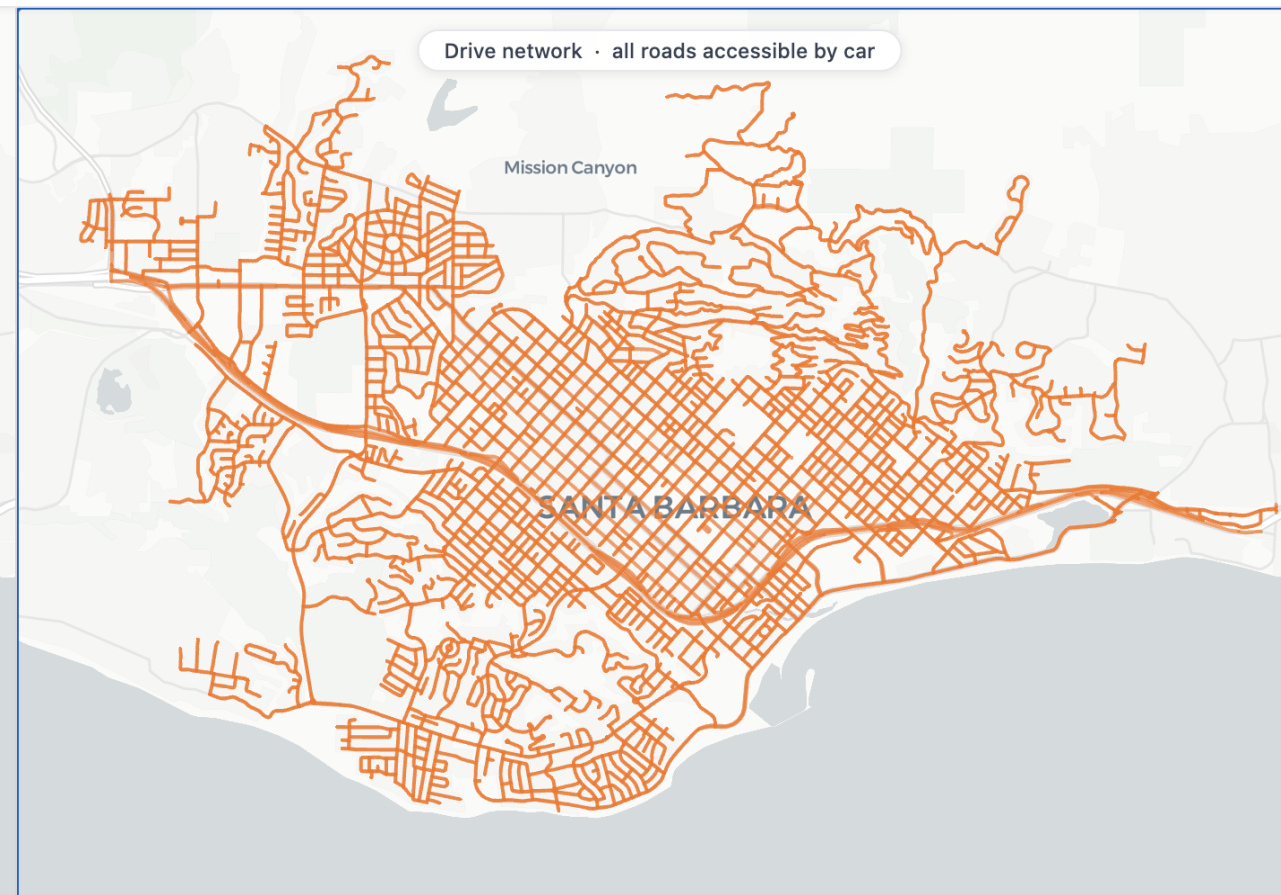
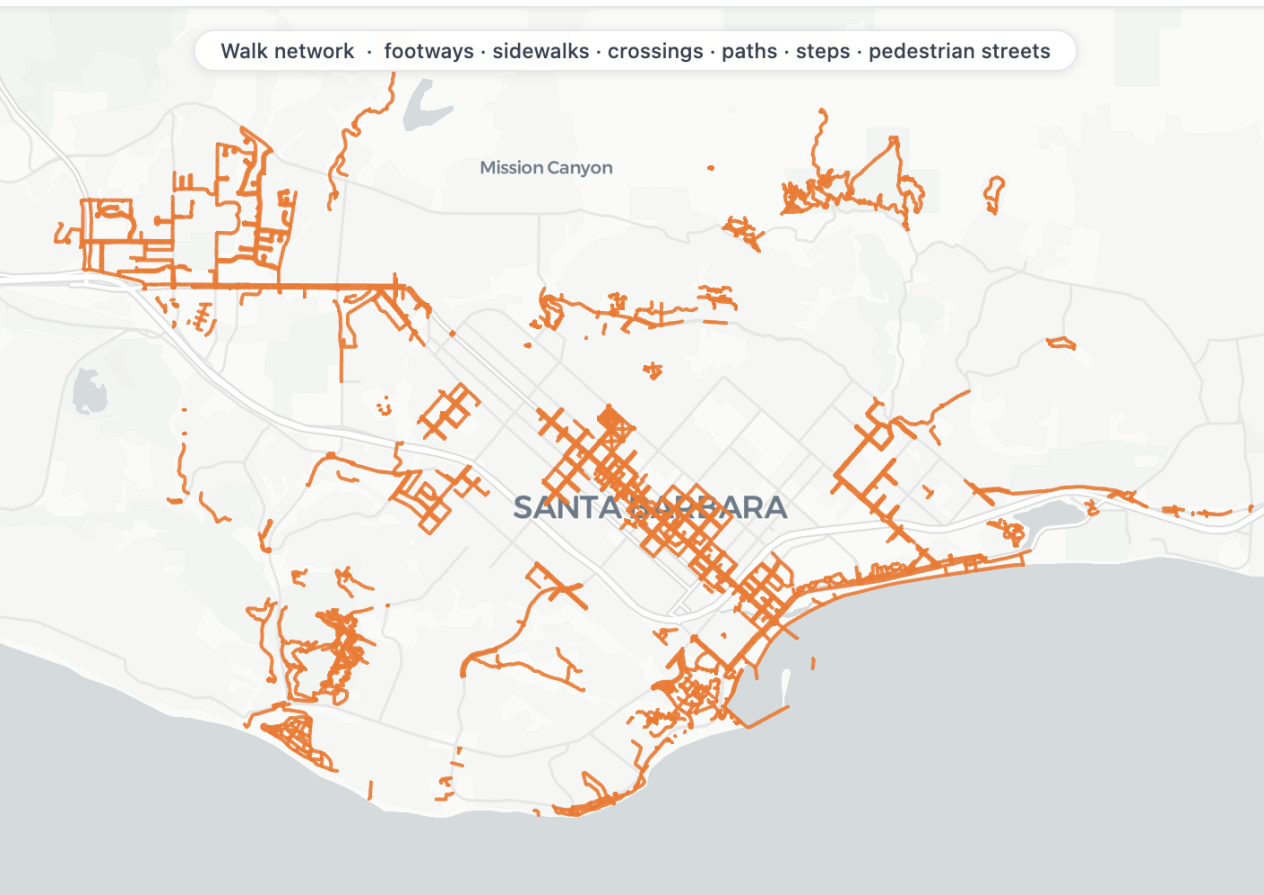
Santa Barbara

Walk Network

Road Network

Santa Barbara — Network edges

Scroll to zoom · Drag to pan · Maps are synced



Davis

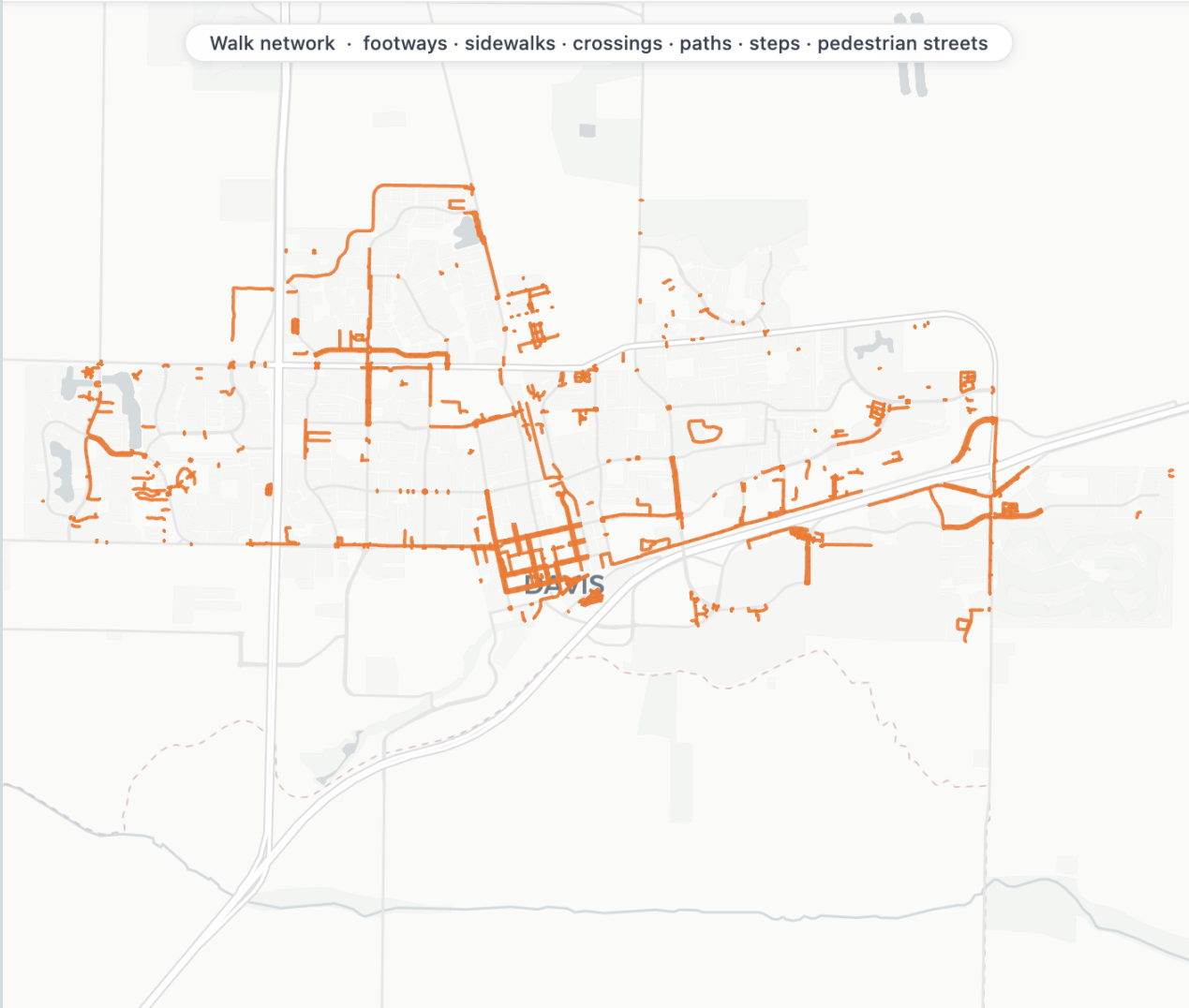
Walk Network

Road Network

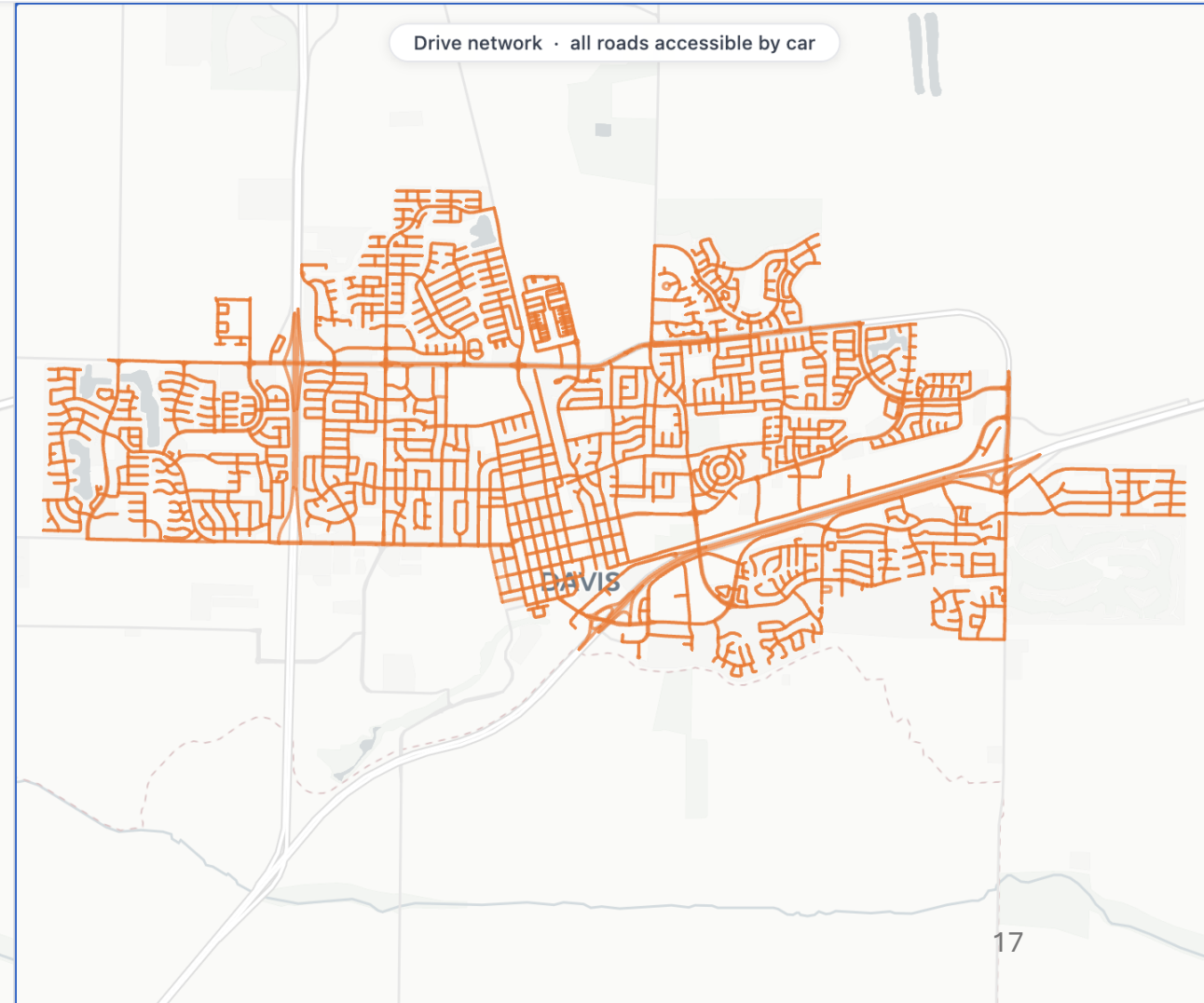
Davis — Network edges

Scroll to zoom · Drag to pan · Maps are synced

Walk network · footways · sidewalks · crossings · paths · steps · pedestrian streets



Drive network · all roads accessible by car



How to Read a Paper?

- You have **8 minutes** to read [this paper](#) in the class and take notes.
- You can add your notes to [this board](#).

How to Read a Paper?

- Now read the same paper, this time look for answers to these questions:
 - 1- What distinct meanings of "walkability" does Forsyth identify, and which describe environmental conditions versus outcomes?
 - 2- What other concepts have the overlaps with walkability?
 - 3- Pick one definition: what would you have to measure to test it, and what would that measure miss?